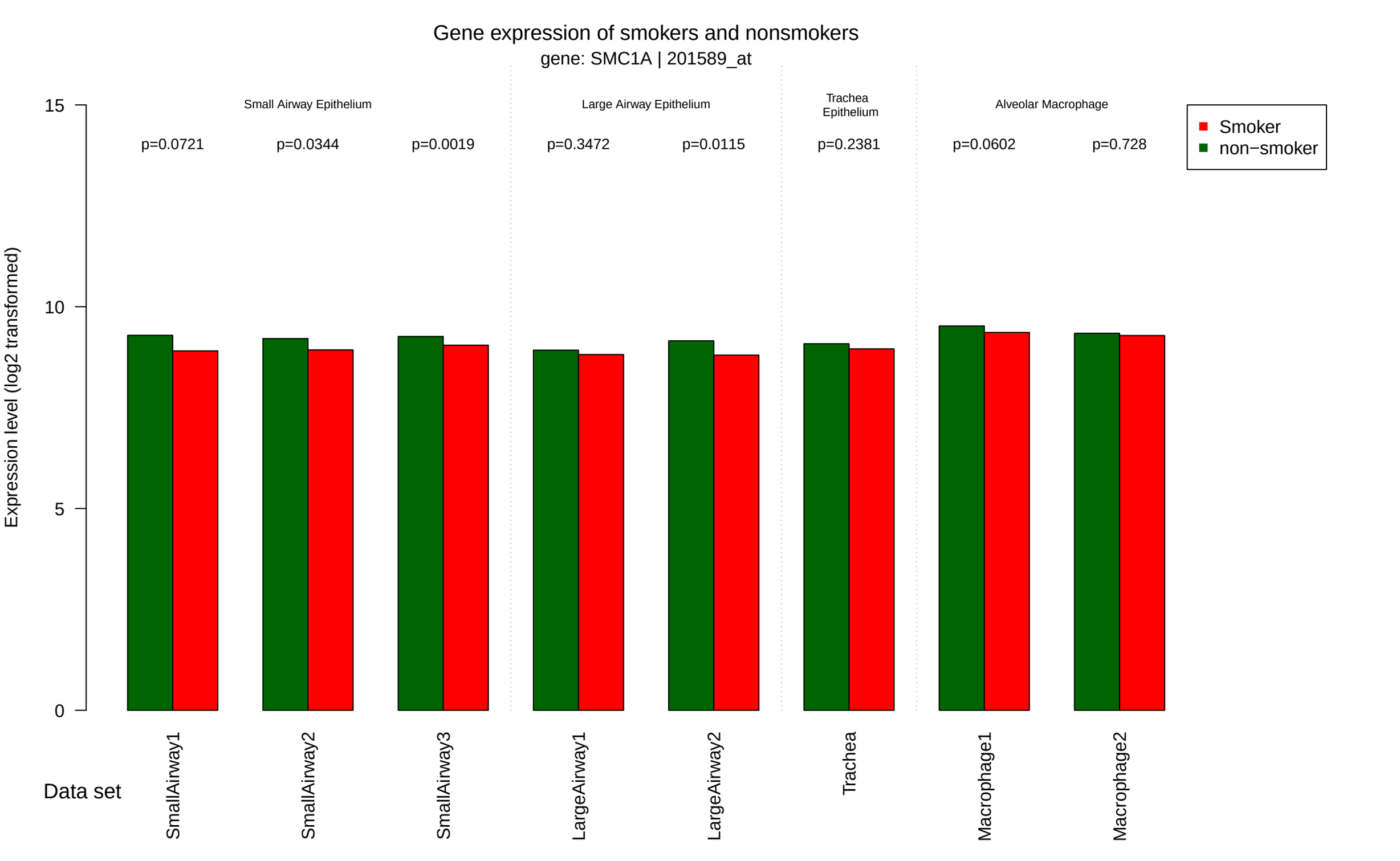
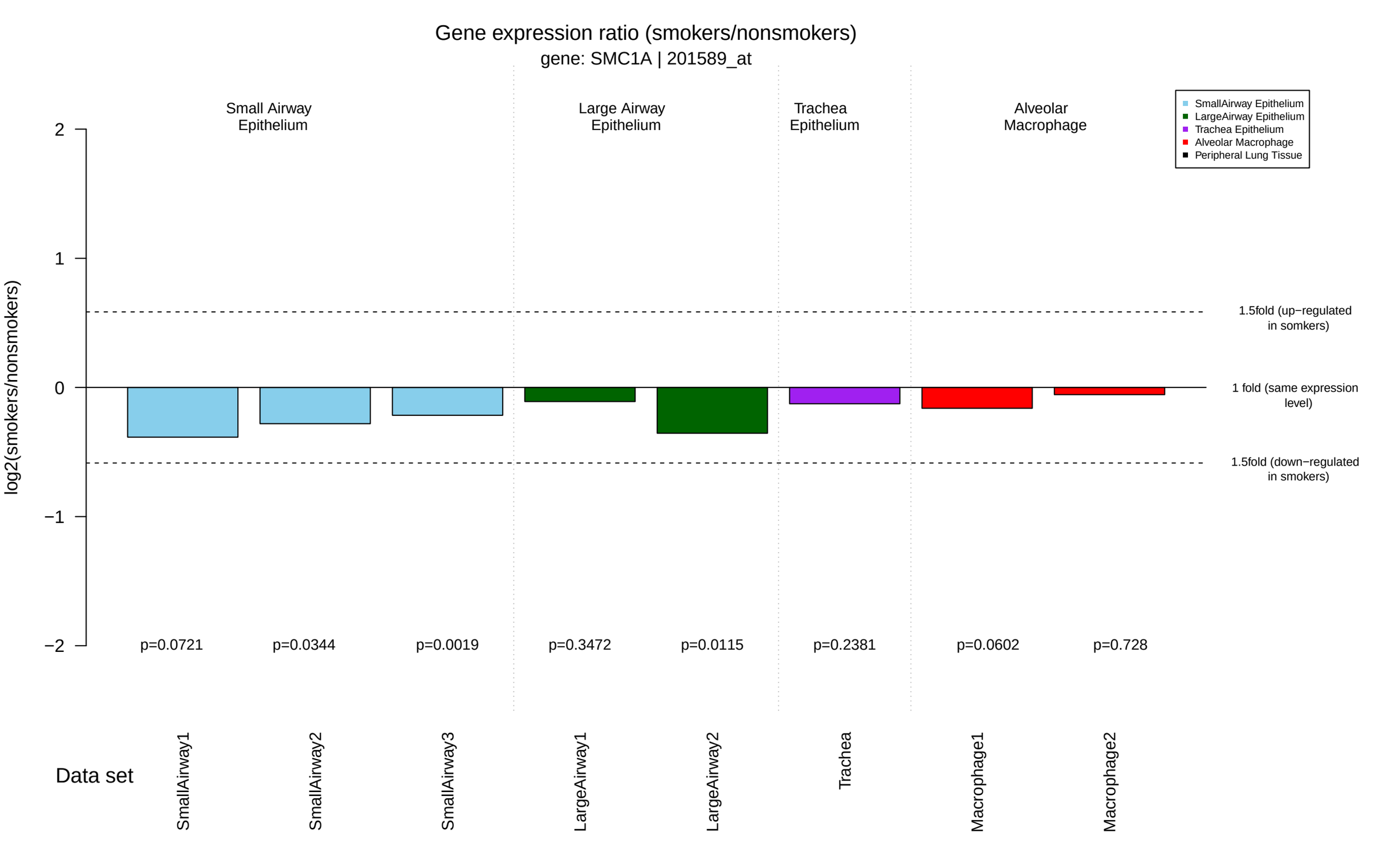
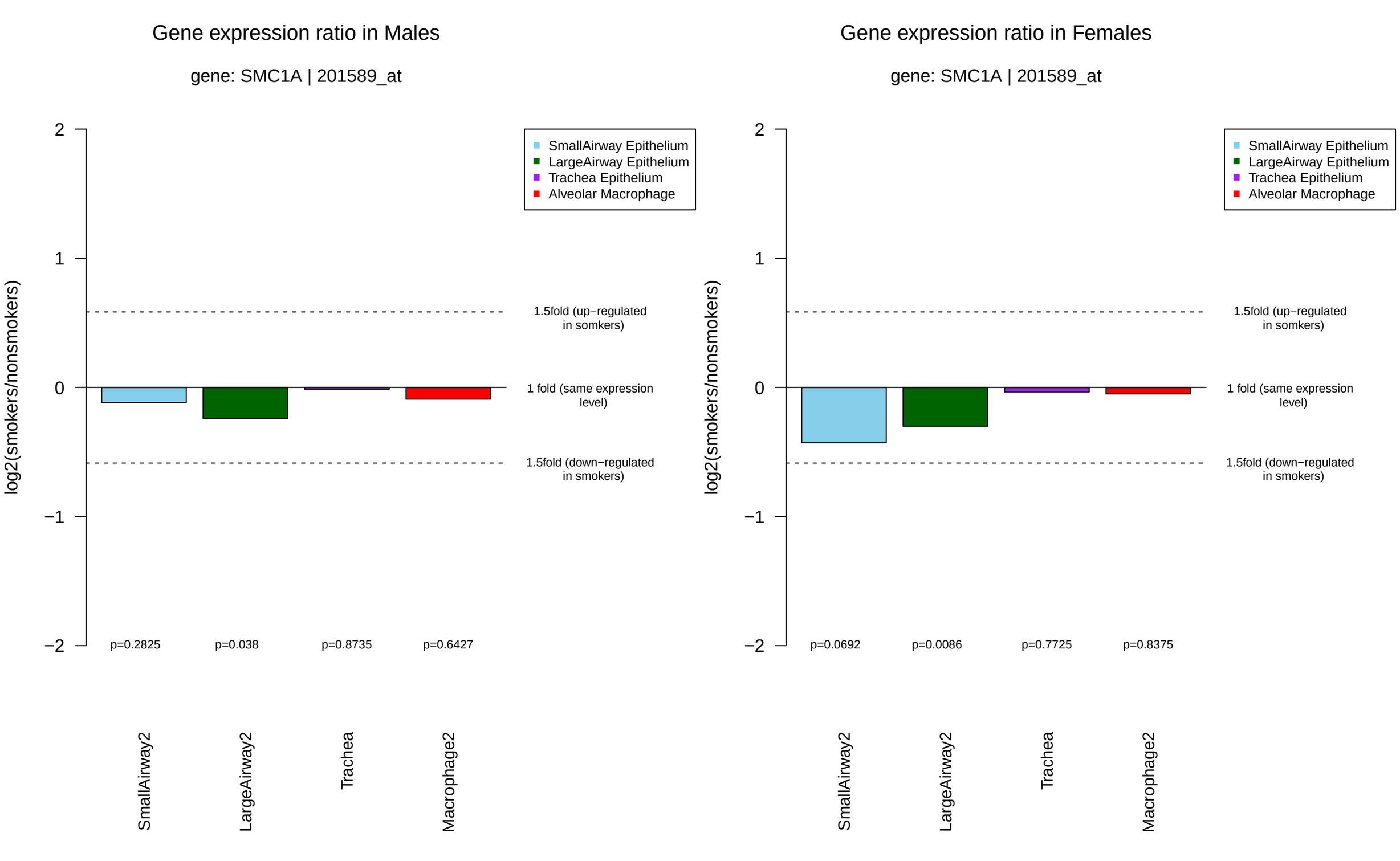
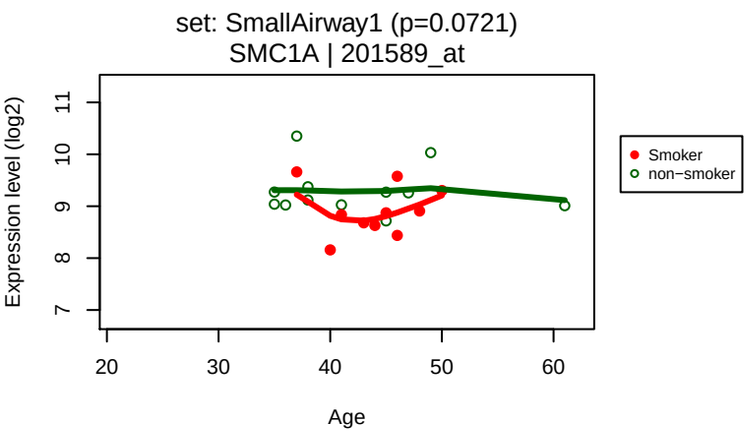
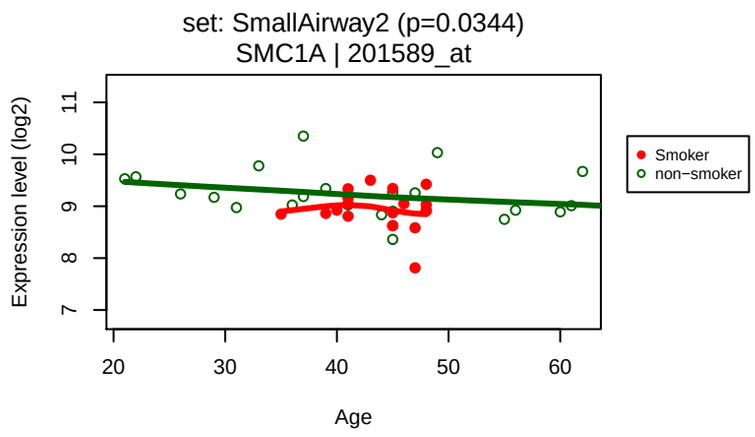
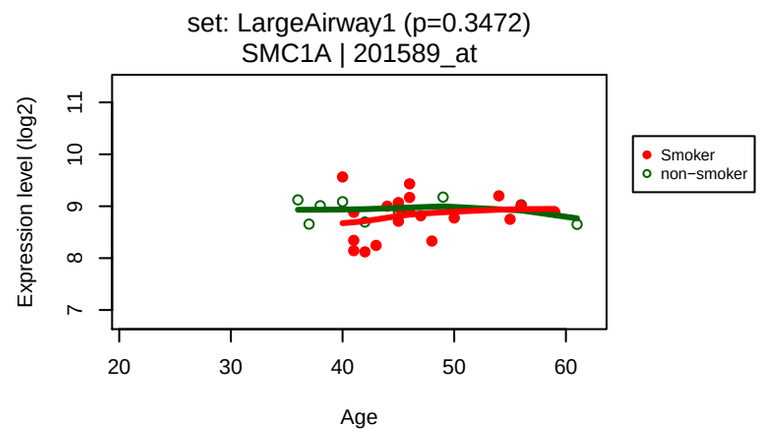

Gene expression of smokers and nonsmokers
gene: SMC1A | 201589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0721
p=0.0344
p=0.0019
p=0.3472
p=0.0115
p=0.2381
p=0.0602
p=0.728
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

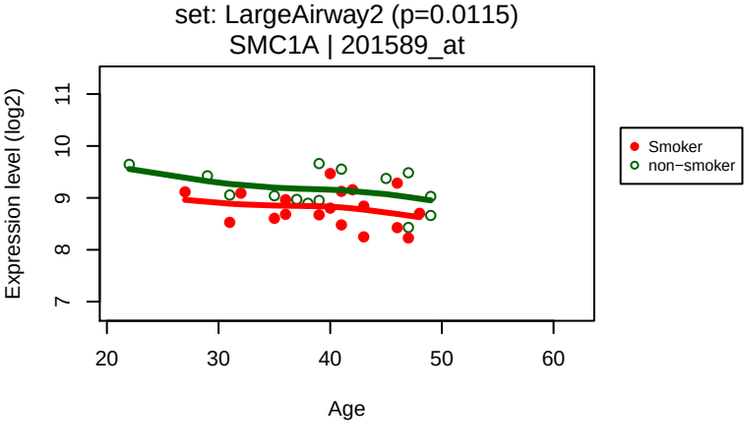

Gene expression ratio (smokers/nonsmokers)
gene: SMC1A | 201589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0721
p=0.0344
p=0.0019
p=0.3472
p=0.0115
p=0.2381
p=0.0602
p=0.728
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

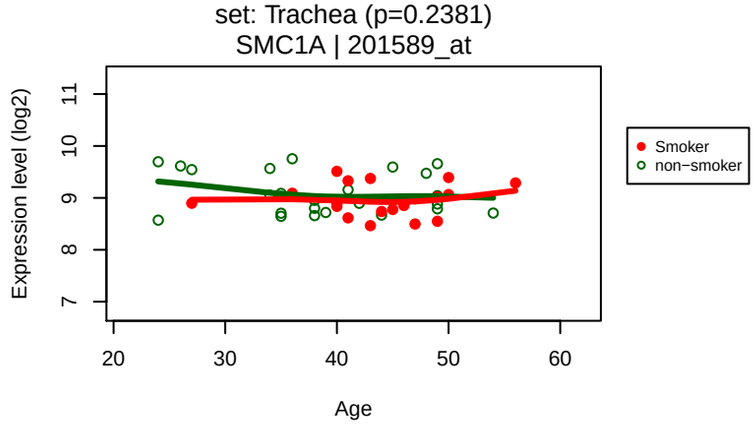

Gene expression ratio in Males
gene: SMC1A | 201589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2825
p=0.038
p=0.8735
p=0.6427
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMC1A | 201589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0692
p=0.0086
p=0.7725
p=0.8375
SmallAirway2
LargeAirway2
Trachea
Macrophage2

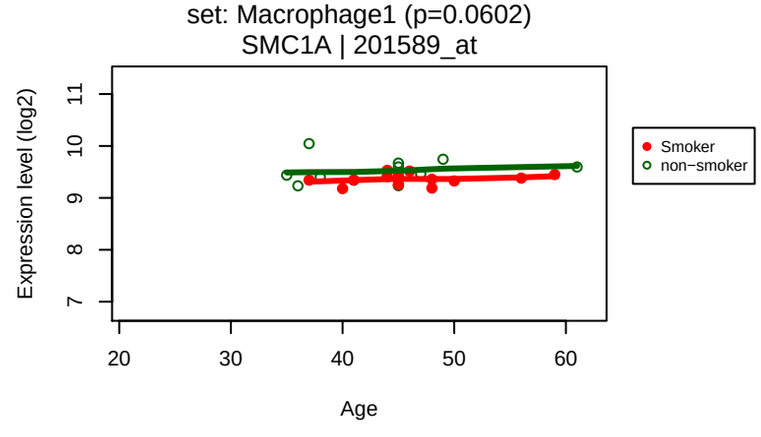

set: SmallAirway1 (p=0.0721)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker

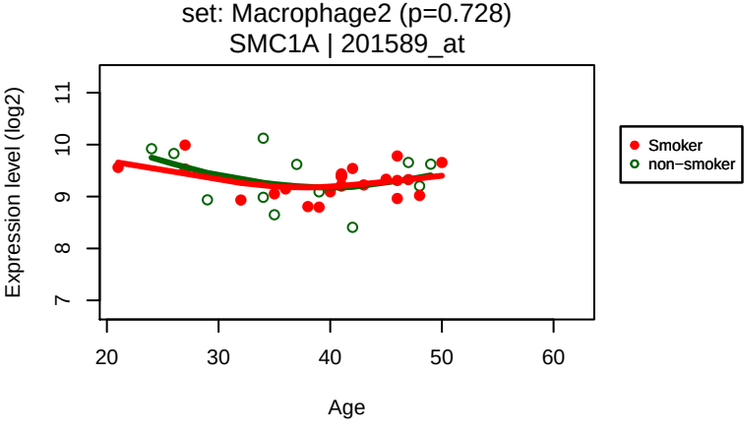

set: SmallAirway2 (p=0.0344)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker

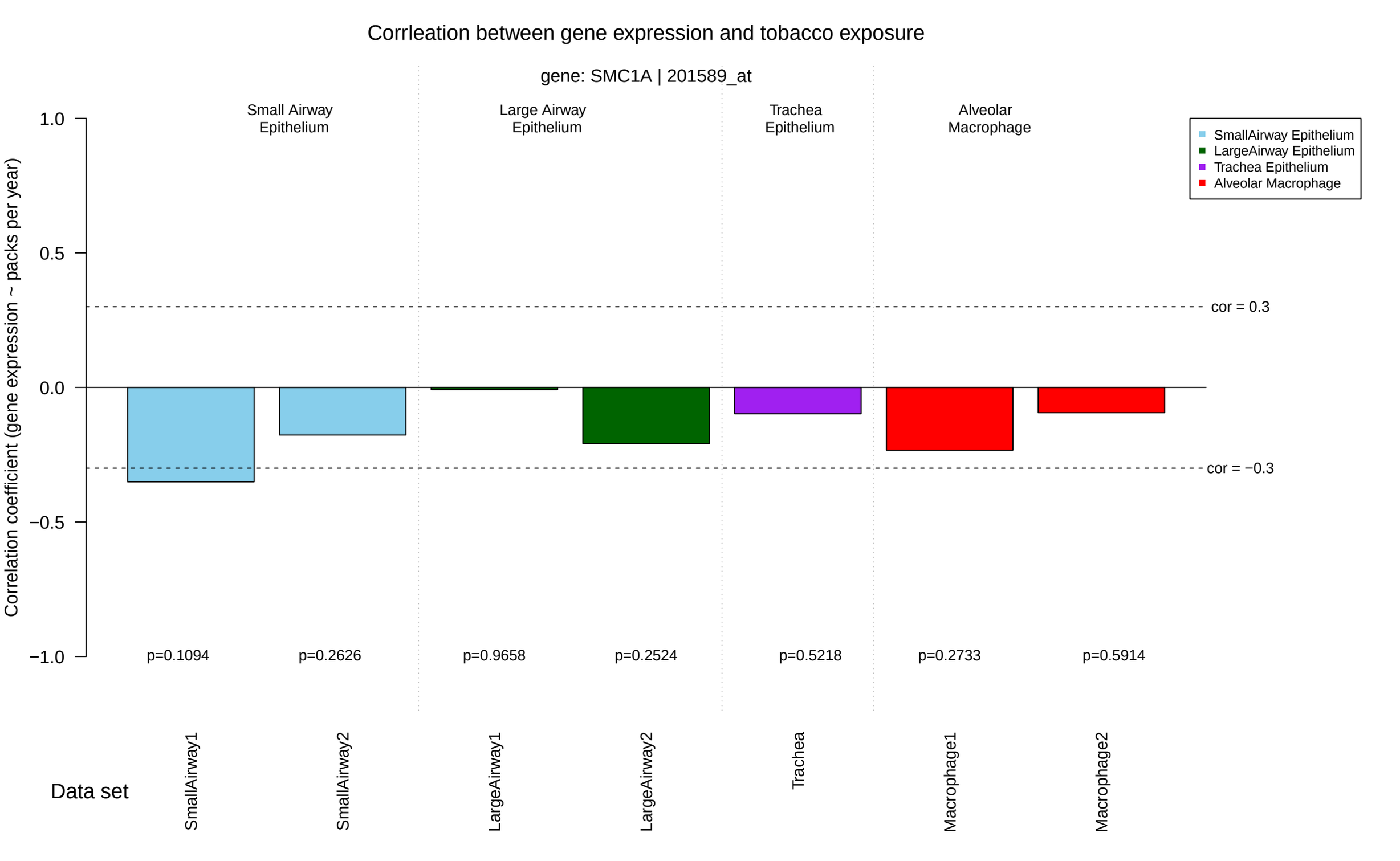

set: LargeAirway1 (p=0.3472)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0115)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2381)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0602)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.728)
SMC1A | 201589_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMC1A | 201589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1094
p=0.2626
p=0.9658
p=0.2524
p=0.5218
p=0.2733
p=0.5914
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2